Borneo HackWknd 2026
The Role of AI in ASEAN Social Impact

Horizon Team
Project: AI-Driven Indigenous Language Preservation Platform

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1.0 Introduction

This project focuses on building a platform that uses artificial intelligence to preserve and revitalize indigenous languages in ASEAN. The app is designed to lower the barrier to language preservation by allowing indigenous language to be recorded, organized and transformed into simple learning content, even when access to native speakers is limited. Instead of relying on large data sets or complex infrastructure, the project demonstrates how AI can support gradual, community-driven language preservation, starting from small and realistic inputs.

The platform primarily support SDG 4: Quality Education by providing accessible AI- assisted learning for endangered indigenous languages. By using artificial intelligence to provide personalized lessons, pronunciation feedback, and adaptive learning path, the platform enables learners to gradually build proficiency even when native speakers or formal teaching resources are limited. In doing so, the platform directly aligned with SDG 4's goal of ensuring inclusive and quality learning opportunities for all.

Many indigenous languages in ASEAN are at risk of disappearing, due to a lack of documentation, limited educational resources and reduced intergenerational transmission. In many communities, indigenous communities are primarily passed down orally by elders. However, younger generations often do not have regular access to these elders, and there are few digital tools that support indigenous languages preservation and learning. As a result, valuable linguistic and cultural knowledge is gradually lost, and existing learning-language platforms do not adequately address the needs of minority and indigenous communities.

The main objectives of the project are to increase access to indigenous language learning, which provides an accessible digital platform that allows users to learn indigenous languages anytime and anywhere, especially in communities where formal learning resources are limited. Next, encourage community participation in language preservation, where the platform is curated by native speakers, elders and language champions. Lastly, leverage artificial intelligence for language revitalization which demonstrates how AI technologies can support language preservation by assisting with speech recognition, translation, and interactive learning experiences.

2.0 Review of Existing Solutions

There are several initiatives that have been developed to learn indigenous languages in the ASEAN region, such as Ling App, uTalk, and Memrise. These solutions provide useful features such as using audio from native speakers, accessing indigenous dictionaries, and communicate directly with locals. However, many existing platforms lack personalized learning pathways or strong AI support to guide learners progressively. Additionally, some of them do not offer continuous feedback to support learners' progress especially in their pronunciation. These gaps highlight the need for a more integrated solution that combines AI-driven adaptive learning with community contributions, enabling gradual language acquisition while empowering indigenous communities to actively preserve and share their linguistic heritage.

3.0 Methodology

This project focuses on developing an AI-driven mobile application for indigenous language learning and preservation. The implementation begins by designing a simple user flow where users select a region and language, explore concept-based vocabulary categories, listen to native audio recordings, and practice pronunciation. Artificial intelligence is integrated to support pronunciation analysis and provide continuous feedback to learners, helping them improve gradually without requiring constant guidance from native speakers.

In addition, the platform incorporates a community-driven contribution system where native speakers can submit voice recordings and suggest new words to expand the language database. AI operates in the background to organize pronunciation data and detect inconsistencies, supporting community validation rather than replacing human input. Through this approach, the project demonstrates how AI and community participation can work together to support sustainable indigenous language preservation and learning.

4.0 Results

The development of the AI-driven indigenous language learning platform demonstrates that technology can effectively support language preservation and education. The platform can integrate AI features such as adaptive learning, pronunciation assistance, and interactive exercises to help users gradually learn indigenous languages.

Overall, the platform successfully met its main objectives as it helps increase access to indigenous language learning, encourage community participation in language preservation, and leverage artificial intelligence for language revitalization. By combining technology with community contributions, the platform demonstrates a practical approach to promoting indigenous language learning while supporting broader goals of inclusive and culturally relevant education.

5.0 Prototype Showcase

Figure 1 shows the welcome screen to greet the user and introduce them to the app.

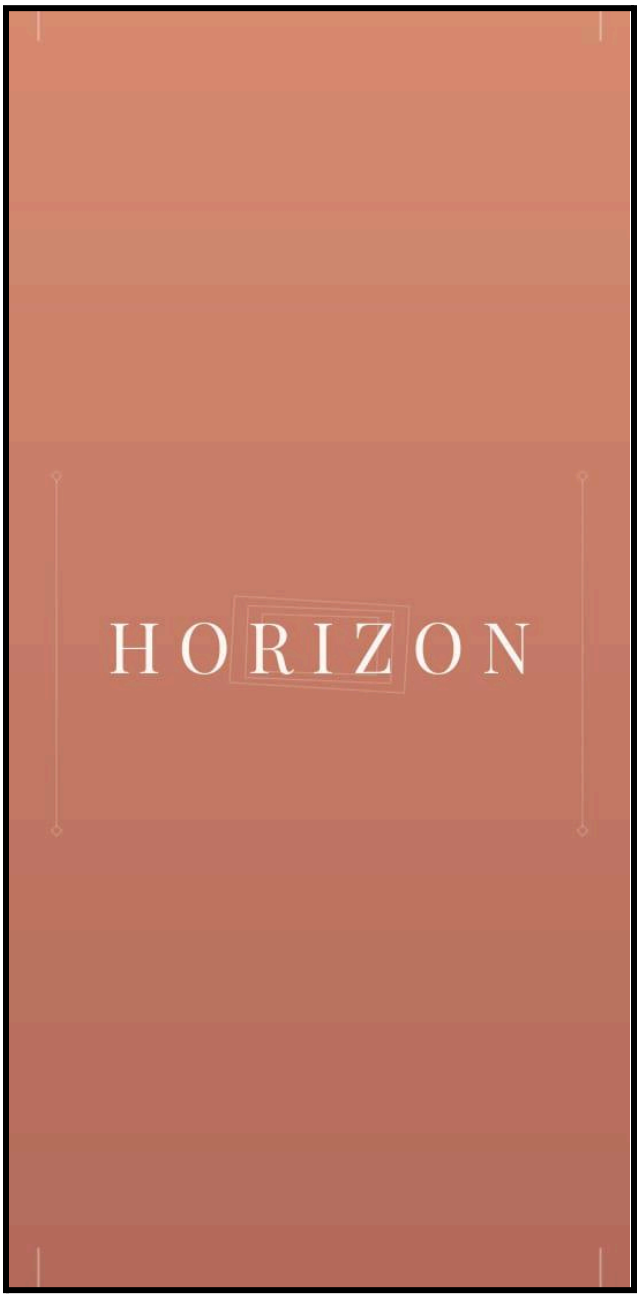

Figure 1

Figure 2 shows the area of regions in Malaysia. Users need to choose a region to learn about their indigenous language.

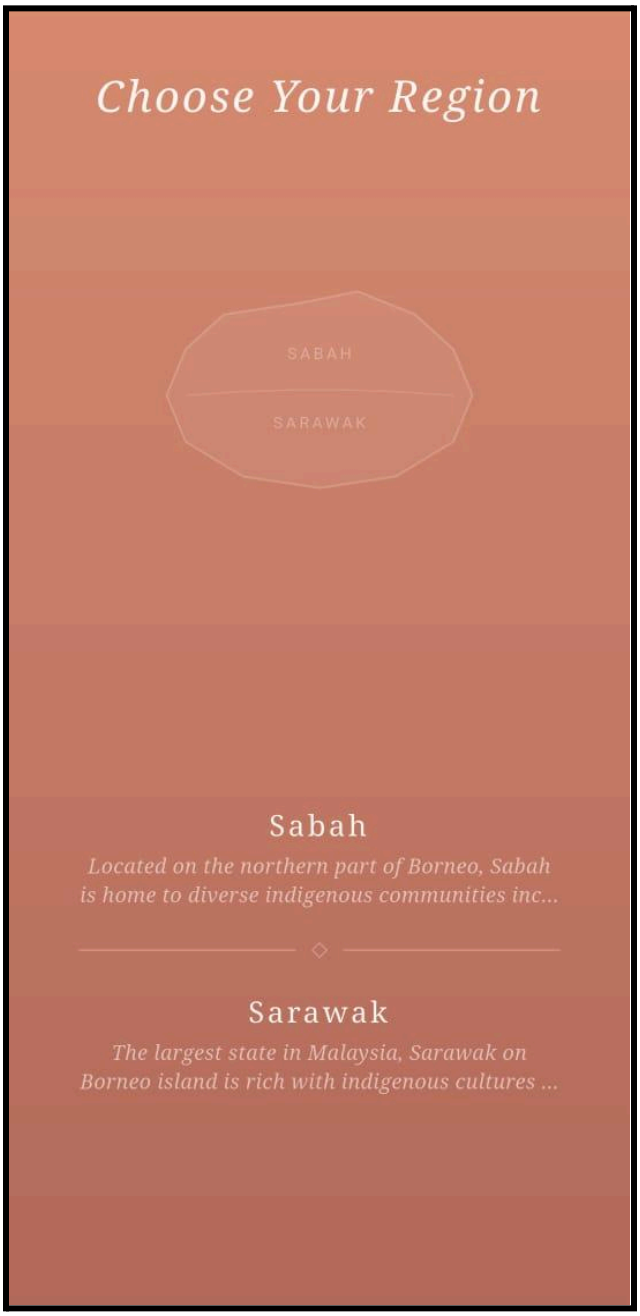

Figure 2

Figure 3 shows a list of indigenous languages selection. Users need to choose one language of the region.

Figure 3

Figure 4 shows a screen asking the user which level are they for the language chosen.

Figure 4

Figure 6 shows the list of topic selection that users would like to learn.

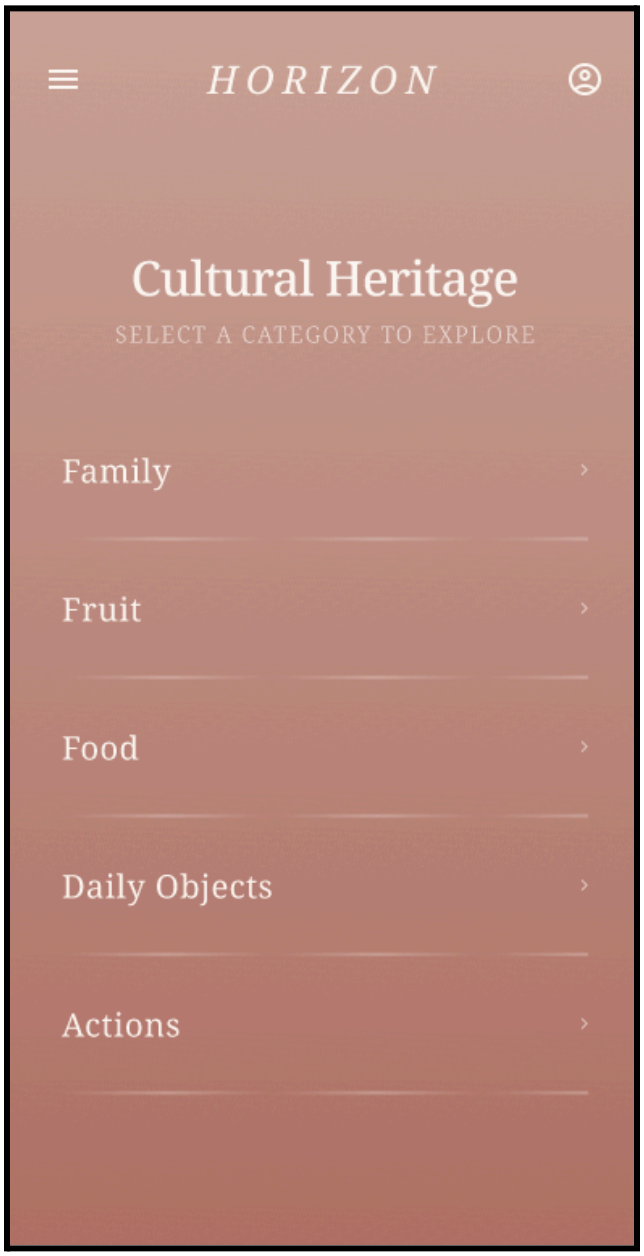

Figure 6

Figure 7 shows the word of the indigenous language and it's translation in English.

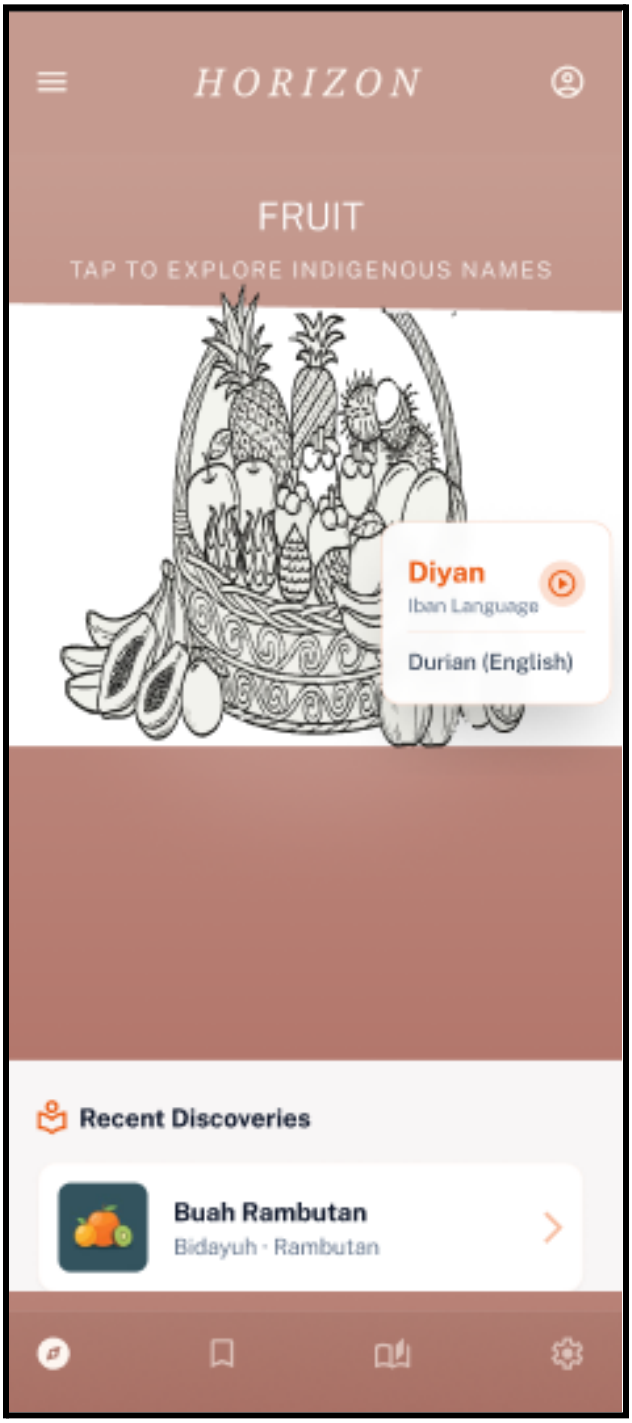

Figure 7

Figure 8 shows the score of the learner's pronunciation accuracy. Learner can choose whether to try again or move to the next word.

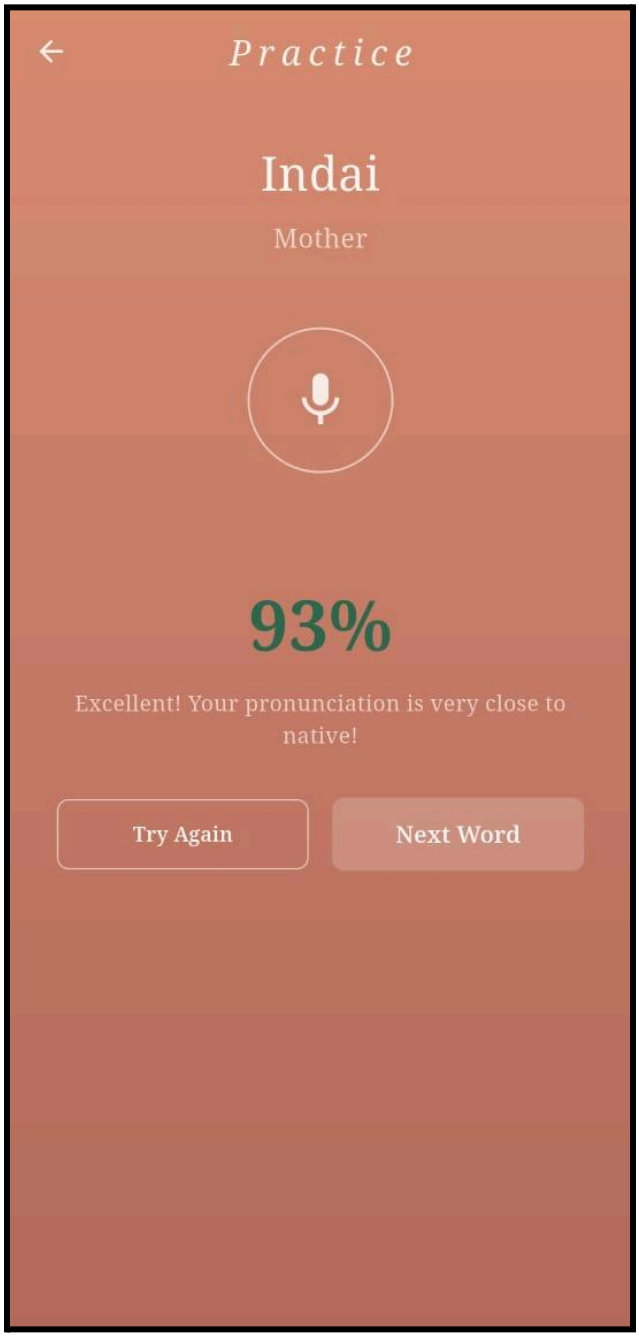

Figure 8

Figure 9 shows a screen asking if the user needs guideline.

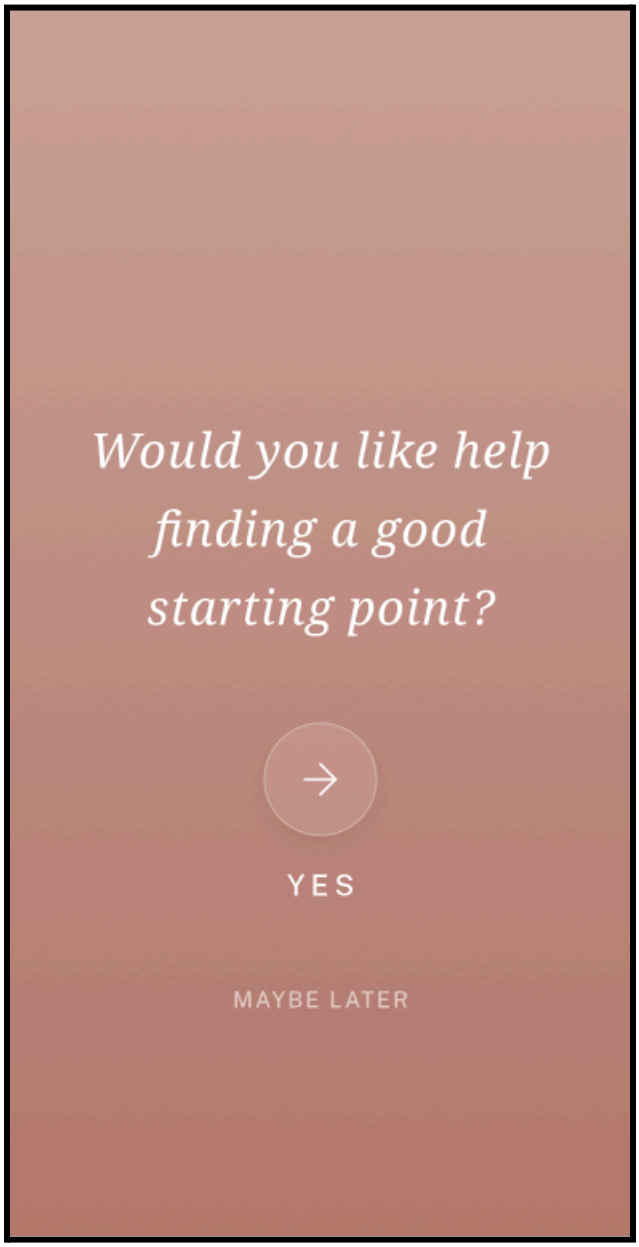

Figure 9

Figure 10 and 11 show the quizzes section to test users' knowledge.

Figure 10

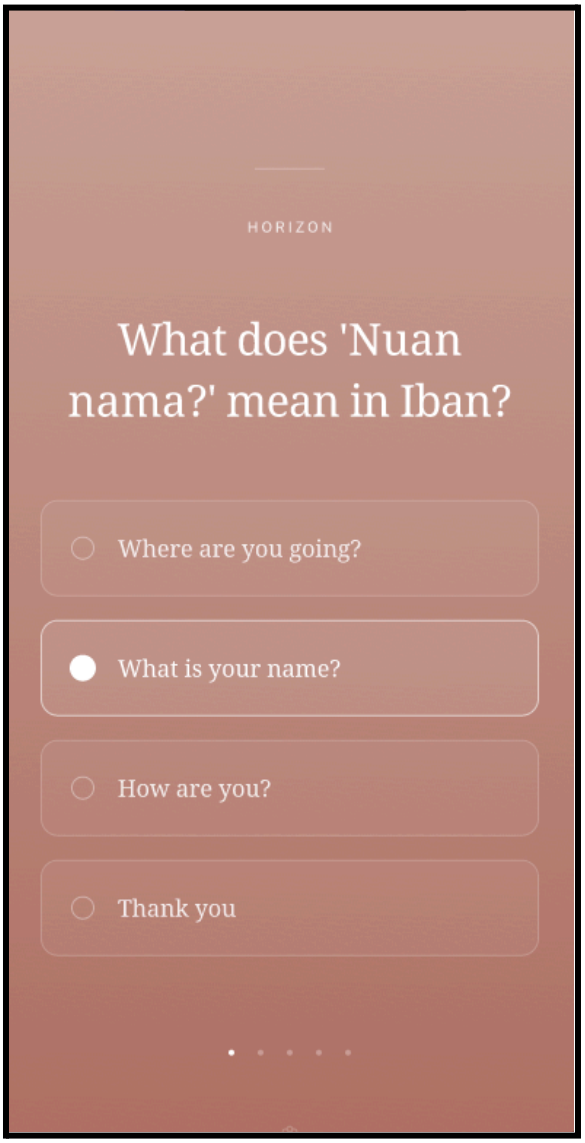

Figure 11

Figure 12 shows user 's personalized learning goal for each day.

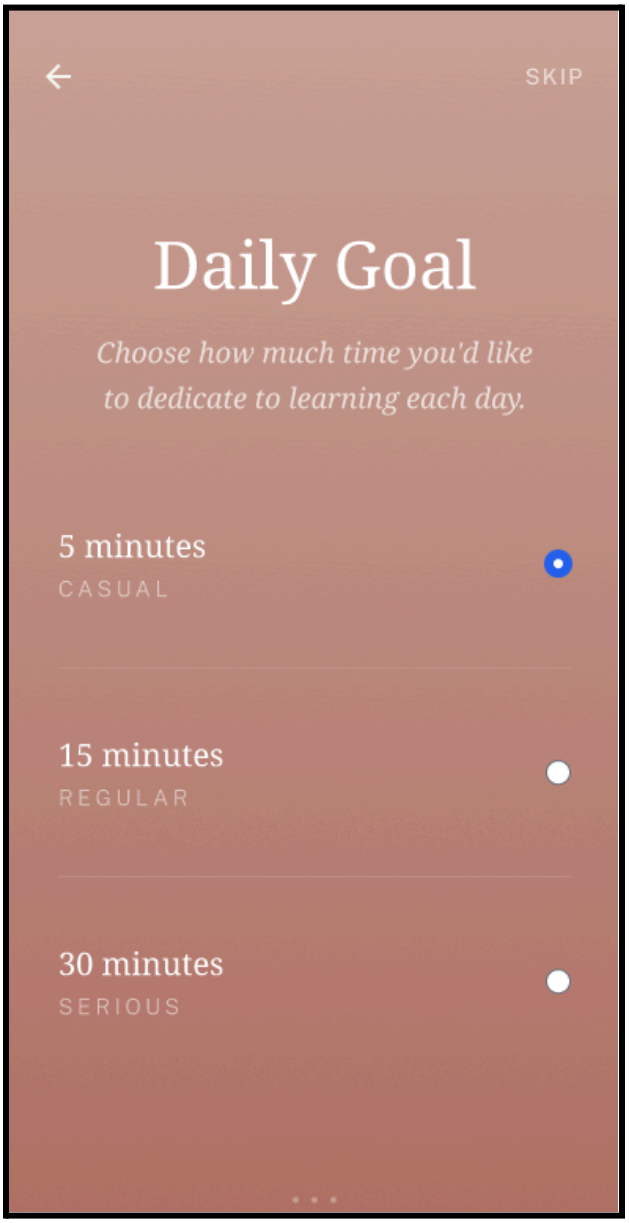

Figure 12

6.0 AI Acknowledgement

Several Artificial Intelligence tools were used during the development and documentation of this project. ChatGPT was used to assist in idea development, language refinement, and improving the clarity of written explanations in the report. In addition, AI-related technologies such as speech recognition (speech-to-text) and audio pattern analysis were conceptually incorporated into the system design to support pronunciation feedback and detect inconsistencies in indigenous language recordings. These technologies enable the platform to convert spoken language into text and assist in identifying potential pronunciation variations so it will make it more accurate.

7.0 Conclusion

In conclusion, this project demonstrates how artificial intelligence can support indigenous language preservation and learning through a community-driven digital platform. By integrating features such as speech-to-text technology, pronunciation analysis, and interactive learning activities, the platform provides a practical approach to documenting and teaching indigenous languages. The platform lowers the barriers to language preservation by allowing native speakers to contribute recordings while enabling learners to access structured learning content and practice pronunciation independently. Through this approach, the project highlights the potential of technology to support cultural preservation while promoting inclusive and accessible language education.

For future improvements, the platform could be expanded to include more indigenous languages across different ASEAN regions and incorporate more advanced AI capabilities for more accurate pronunciation feedback and language analysis. Additional features such as gamified learning, offline learning support for remote communities, and stronger collaboration tools for researchers could further enhance the effectiveness of the platform. With continuous development and community participation, the platform has the potential to become a sustainable digital resource that contributes to long-term indigenous language revitalization and cultural preservation.